Central Queensland University

CQU Rockhampton TAFE Consolidation - Stage 2

Ecologically Sustainable Development (ESD) Report

February 2026

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CQU Rockhampton TAFE Consolidation - Stage 2
Ecologically Sustainable Development (ESD) Report

Central Queensland University

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WSP acknowledges that every project we work on takes place on First Peoples lands.
We recognise Aboriginal and Torres Strait Islander Peoples as the first scientists and engineers and pay our respects to Elders past and present.

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1 Introduction

This Report is provided by WSP Australia Pty Limited (*WSP*) for Central Queensland University (CQU) Australia (*Client*) to support the sustainable delivery of the New Trades Building, and refurbishment of Building 33 and 36, located in the CQU Rockhampton North campus.

1.1 Project description

New Trades Building

The proposed Trades Building is a two-storey building, located west of the existing Building 70 (Centre for Railway Engineering) on the CQU Rockhampton North Campus. The New Trades Building covers the same footprint as existing Buildings 24, 23 and their associated carparks. Refer to Figure 1.1 and Figure 1.2 for figures of the site location and proposed development layout.

The New Trades Building will house multiple industrial training areas, including light automotive and auto electrics, metal fabrication, fitting and machining and refrigeration. The design prioritises the specific workshop needs of each discipline while maintaining proximity to encourage collaboration and operational efficiency. Shared storage facilities, flexible theory rooms, and bookable spaces have been incorporated to optimise resource use and reduce unnecessary duplication of infrastructure.

Building 33 & 36 Refurbishment

Building 33 and Building 36 have been identified for significant refurbishment works. Building 33 will undergo a comprehensive upgrade to two levels of its existing three-level storey structure, transforming it into teaching and administrative spaces to support flexible learning environments. Building 36 will be fully refurbished to accommodate specialised Tafe faculties, incorporating public-facing services to meet comprehensive training needs.

Given the extend of these refurbishments both have been included in the ESD targets for the project. Buildings 19, 32 and 34 are disregarded in accordance with the CQU Sustainability Guidelines due to their limited scope of works, further detail is noted in Sections 2.1 and 4. Refer to Figure 1.2 for an overall staging plan and campus map.

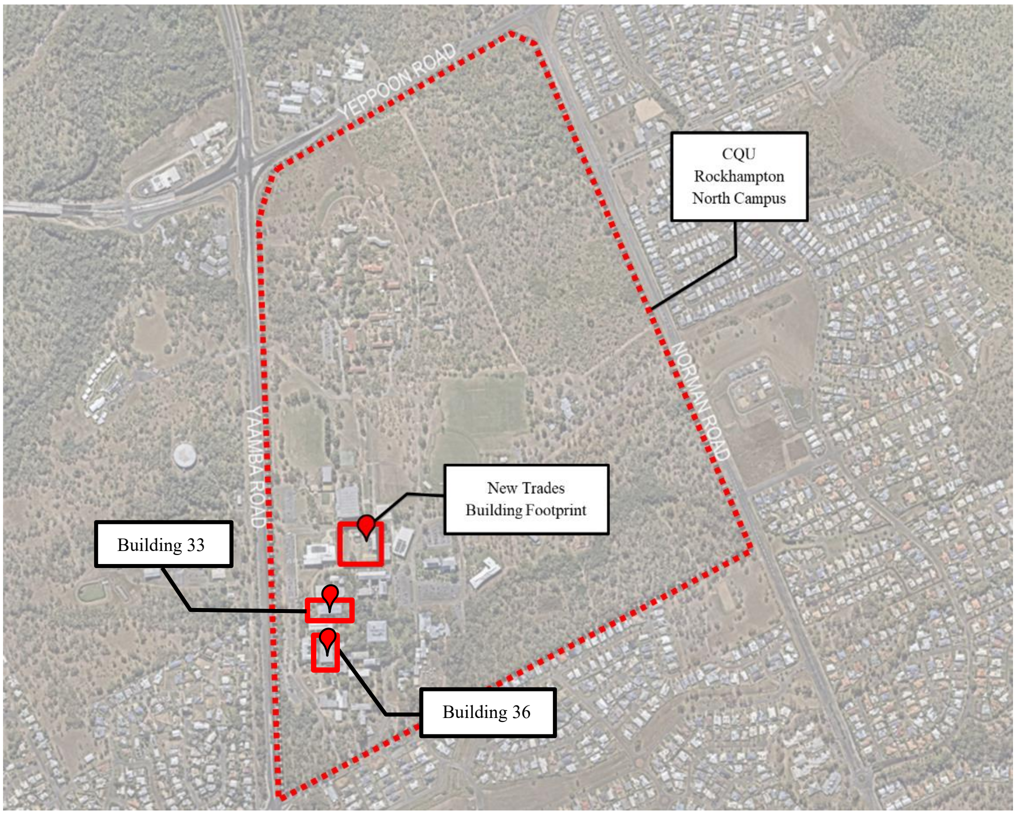


Figure 1.1 CQU Rockhampton North Campus Site Map (Source: Nearmap Imagery 2025)

Figure 1.2 CQU Rockhampton – Architectural Design Report (Rev C) by Peddle Thorp.

1.2 ESD Scope

The scope of this document is to outline a consolidated design sustainability plan for the Rockhampton CQU Tafe campus new Trades Building development and refurbishment of Building 33 and 36 . The role of this report is to provide the necessary advice to the project team on achieving best practice, and exemplary sustainability targets associated with the Green Star (GS) Building V1 targets for both the new trades building and refurbishments. Whilst CQU requires designs to meet the relevant Green Star criteria, attaining a formal Green Star Certification is not being targeted for this project. A holistic approach will be considered to ensure that the facility incorporates the sustainability commitments and frameworks of CQU and their correlating 2024 – 2028 Strategic Plans. Refer to Sections 3 & 4 for an overview of the Sustainability Strategy and Green Star Buildings pathways for the proposed development.

1.3 ESD Targets

The proposed development will target the following sustainable initiatives:

- 5-star Green Star Buildings V1 design equivalence for the new Trades Building
- 4-star Green Star Buildings V1 design equivalence for the two refurbish buildings (33 and 36).
- Compliance with NCC Section J 2022 Building Fabric requirements

1.4 Sources of Information

The following documents has been used in the preparation of this report:

- Queensland Development Code (QDC), MP 4.1 – Sustainable Buildings, Version 1.15, Issued 19 September 2023.
- Green Star Buildings Submission Guidelines Version 1.1 Revision C 18th October 2023.
- National Construction Code (NCC) 2022 Volume One, Section J Energy Efficiency.
- Architectural DD Drawing Set from Peddle Thorp, issued December 2025.
- CQUniversity Sustainability Policy 2024
- CQUniversity Annual Sustainability Report 2025
- CQUniversity Sustainability Framework 2024 – 2028 | Guiding our sustainability journey
- CQUniversity Strategic Plan 2024 – 2028
- CQUniversity Decarbonisation Strategic Plan 2024 – 2028

2 ESD Policy and Drivers

Several sustainability frameworks exist at a global, federal, state and local level that have been used to provide the context for goals, objectives and targets for the ESD approach for the project. These include:

- United Nations Sustainable Development Goals (Global).
- The Paris Agreement (Global):
  - Under the Paris Agreement, Australia has committed to reducing emissions by 26-28% on 2005 levels by 2030 (Federal).
- National Construction Code 2022 Section J Compliance (Federal).
- Queensland Development Code (QDC) MP 4.1 – Sustainable Buildings (State).
- Queensland Net Zero Roadmap & Net Zero by 2050 Legislation (State).
- Queensland Transport Strategy (State).
- Rockhampton Regional Council Sustainability Strategy & Policies. (State)
- CQUniversity Sustainability Policy 2024 (local).
- CQUniversity Annual Sustainability Report 2025 (local).
- CQUniversity Sustainability Framework 2024 – 2028 | Guiding our sustainability journey (local).
- CQUniversity Strategic Plan 2024 – 2028 (local).
- CQUniversity Decarbonisation Strategic Plan 2024 – 2028 (local).

2.1 CQU Sustainability guidelines

CQU is committed to embedding sustainability across all aspects of its operations, education, research, and community engagement. Guided by its Sustainability Policy and the 2024-2028 Sustainability Framework, CQU aligns its actions with Environmental, Social and Governance (ESG) and the United Nations Sustainable Development Goals (SDGs). Progress is tracked via annual reporting, measurable targets and key requirements, including:

- Sustainability is built into all university activities – teaching, research, operations, and partnerships, with staff, students and partners encouraged to get involved in sustainability efforts.
- CQU sets clear goals and tracks progress across nine frameworks, such as energy, water, waste, biodiversity, and built environment
- The University is working to cut emissions and use resources efficiently, aiming for net zero by 2050.
- All new buildings must meet high sustainability standards (5 Star Green Star Rating) but do not require GS certification.
- All fully refurbished buildings must meet the 4 Star Green Star Rating (no GS certification required) however any buildings undergoing partial refurbishment do not require any design rating.

3 Sustainable Strategy

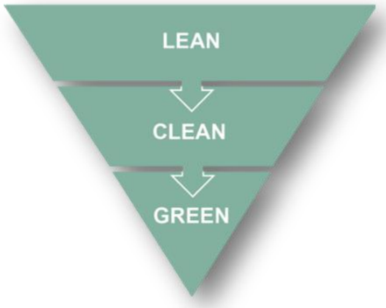
The following section sets out the sustainability strategy for the development and addressees the project's aim to demonstrate a high standard of environmental design, in accordance with the CQU Sustainability Frameworks.

3.1 Energy and greenhouse gas emissions

3.1.1 Energy efficient strategy and targets

The reduction of GHG emissions in the built environment is a major focus to curb the impacts of climate change. Improving energy efficiency leads to a reduction in carbon emissions and reduces the consumption of finite resources.

The strategy for emissions reduction is to follow a "Lean, Clean, Green" approach, balancing immediate environmental and economic performance with long term deep-cut emissions potential. This approach has delivered opportunities to maximise development potential while minimising carbon footprint.



- **Lean** | To mitigate the demand for resources through prioritising passive design strategy.
- **Clean** | Selection of efficient equipment to deliver further improvements in energy and water performance.
- **Green** | Selection of green technology like Photovoltaic (PV) system to reduce carbon emissions.

The current Green Star pathway for the project includes the following targets:

- A minimum 20% lower energy use than one built under the relevant NCC version applicable to the project.
- Eliminate or offset emissions from refrigerants.
- 100% of the energy under the control of the building owner/operator, and all non-electricity energy provided for users that are not under the building owner's control, must be sourced from renewable.
- A minimum 10% reduction in embodied carbon and operational energy use

3.1.2 Energy efficient initiatives

The following initiatives are proposed to ensure the project reduces its carbon emissions as far as possible with on-site measures:

- Integrate building services design: high-efficiency ducted systems provided for localised areas (in workshops, CNC rooms, CAD classrooms ARC Welding, etc) so they can be turned off if unoccupied. All systems managed by efficient pumps all managed central BMS for optimal performance and minimal maintenance.
- High-efficiency appliances: All new electrical appliances will achieve the highest available energy rating at the time of development. This will further be detailed at the Detailed Design stage.
- High-efficiency LED lighting: All lighting with be LED, meeting IEEE 1789-2015 standards for mitigating flickering and health risks to viewers, with a minimum Colour Rating Index (CRI) of 85 for all internal and external applications. Lighting will also include smart controls such as dimmers, motion detectors and automatic switches.
- The project will use Water Sensitive Urban Design by harvesting rainwater for toilets and irrigation, installing efficient fixtures, using drought-tolerant landscaping, and including smart irrigation controls and stormwater treatment features to reduce water and energy use while protecting local waterways.

3.1.3 Energy metering and monitoring

Electrical sub-metering is to be provided for significant end uses such as lighting, refrigeration, MSSB/HVAC and each trade department rooms. Metering of energy consumption can assist considerably in ensuring that energy used in operation is measured and monitored so that it can be reduced.

The meters will be connected to a central BMS or similar logging system, capable of producing hourly, daily, monthly and annual reports for effective monitoring. A monitoring strategy will be developed, including energy targets to track operational usage.

3.1.4 Generation and Storage of Renewable Energy

Due to the large roof space of the new two-storey facility, approximately 50kW of rooftop solar panels are proposed. The PV system will be integrated with building management controls to optimise self-consumption and support critical loads.

As this is an educational building, it is expected that the peak demand of the building will occur during daytime teaching hours, which aligns with the period of maximum solar PV generation. Therefore, battery storage is not proposed as most of the solar energy produced can be used directly to offset daytime loads.

3.2 Water consumption

The water strategy for the building will be to first reduce consumption through maximised efficiency. The next step will be to include metering and monitoring to identify any leaks or unusual uses.

To achieve water efficiency there will be a heavy emphasis on the efficiency of the water fixtures and fittings, however still maintaining an acceptable level of flow. All new products (taps, toilets and dishwashers) will achieve the maximum WELS rating available at the time of installation to comply with the CQU frameworks. Minimum requirements mentioned regarding the Green Star minimum requirement as outlined in Table 3.1 below:

Table 3.1 Recommended water fixtures and fittings efficiencies

Fixture	WELS Rating
Toilets	min 4 Star – 3.5L/flush
Urinals	min 5 Star
Taps	min 5 Star - 6L/min
Showers	min 3 Star – 7.5L/min
Dishwashers	Highest star rating (min 5 Star)
Washing Machines	Highest star rating (min 4 Star)

Other water efficiency measures for the development include:

- Use of native vegetation as opposed to exotic species in applicable areas, which encourage native wildlife and have lower water requirements.
- Use of a water efficient irrigation system, such as drip irrigation, to reduce water consumption.
- A 30kL rainwater tank is proposed only as part of the new Trades Building, to supply non-potable uses such as toilets and campus irrigation.

3.3 Materials use

3.3.1 Sustainable material targets

Where appropriate, priority will be given to procuring materials with a low environmental impact, are recycled or contain recycled content, are locally sourced and are durable with low maintenance. This is to help achieve the following initiatives outlined for the project.

- To reduce building occupants being directly exposed to toxins the building will specify the use of low VOC paints, adhesives, sealants and carpets, and low formaldehyde engineered wood products within occupied spaces. This includes prioritising using toxin free flooring and using mechanical fittings instead of adhesives or glues.
- Where timber is used, priority will be given to timber from plantations or sustainable managed growth forests (sustainability certified timber). Timber will not be sourced from rainforests or old growth.
- Reviewing procurement to improve the efficiency of materials usage, reducing single-use items, and purchasing items that are carbon neutral or with low embodied carbon, such as for renovations and new construction projects, office supplies, copy paper.
- Any new construction or refurbishment is to critically assess and reduce the need for any excess materials including future proof/redundancy items. This can be building materials, furniture, wiring/cabling etc.

3.3.2 Upfront carbon emissions

CQU's Decarbonisation roadmap to 2030 prioritises Scope 1 and 2 emissions, with scope 3 scheduled to commence after 2028. To align with this future focus and Green Star Building materials credits, CQU aims to ensure projects begin with preparing for low-carbon material specifications. Some of which including, but not limited to:

- Low-embodied-carbon concrete (e.g. SCMs), steel and aluminium and PEFC-certified timber where feasible.
- Design for material efficiency (right-sizing structures, standardised grids, prefabrication/modular elements) and waste minimisation (off-site fabrications)
- Select low-VOC finishes to support indoor environmental quality while minimising recurring embodied energy through durable maintainable materials

3.4 Climate adaption pre-screening

As part of the Green Star Buildings Credit 16 requirements and in alignment with the CQU Sustainability Frameworks, the project requires thorough climate change pre-screening and risk review. This process aims to assess potential impacts from heatwaves, extreme rainfall, flooding, wind, bushfire smoke, and supply disruptions.

This process considers both historic and future climate data to identify risks such as damage to building components, accelerated deterioration or reduced design life, and reduced operating capacity. The pre-screening covers impacts to surrounding areas, occupant wellbeing and essential services such as transport, power, water and telecommunications. The pre-screening is summarised in Table 3.2 below, with further climate risk detail in Table 3.3.

Table 3.2 Green Star's Climate Change Pre-Screening Checklist

Pre-Screening item	Applies to project	Has data regarding future climate exposure been reviewed?	Has a risk to the project been identified?	Has a risk treatment been identified?
The project area has previously been impacted by extreme climate events (e.g. storms/tropical cyclones, extreme rainfall and flooding, damaging winds, damaging hail, bushfires, heatwaves, drought, or coastal inundation).	Yes	Yes	Yes	TBC
The project is located in a cyclone zone.	Yes	Yes	Yes	TBC
The is project located in or adjacent to a bushfire prone area.	Yes	Yes	Yes	TBC
The project is located in or adjacent to a flood prone area.	Yes	Yes	Yes	TBC
The project is located at or adjacent to the coastline or tidally influenced waterway.	No	N/A	N/A	N/A
The project will accommodate occupants vulnerable to the impacts of climate extremes (e.g. children, elderly, low mobility, seeking medical treatment).	Yes	N/A	Yes	TBC

Table 3.3 Selection of historic climate events with potential site risks

Disaster category	Historic climate events	Potential Project Site Risks
Flooding & Extreme Rainfall	*Jan 2011*: Fitzroy River flooded, reaching a major peak of 9.20m at Rockhampton Gauge (7km from the project site) and widespread inundation across Central Queensland.	The site is likely susceptible to flooding from the nearby Fitzroy River, as interim flood lines from the Rapid Hazard Assessment indicate that part of the site was within the boundary of the 2011 flood event. However, given the distance between the previous flood peak and the site, riverine flooding is not considered a major priority for site-specific risk management. Although, due to extreme rainfall events, local overland flow and roof/drainage surcharge should be considered in project design.
Cyclones and Storm Tides	*Feb 2015:* Tropical Cyclone Marcia – made landfall near Shoalwater Bay; significant damage recorded at Yeppoon and Rockhampton as the system weakened inland.	Cyclone risks such as wind-borne debris, roof uplift and pro-longed outages should be considered and prepared for since the facility is located within a cyclone zone.
Storms, Hail & Extreme Winds	April 19, 2020: Severe hailstorm hit Rockhampton with hail observed in sizes up to 10cm diameter. Hundreds of roofs and vehicles damaged.	Historical hailstorms in the site region show the project must consider the roof and skylight failure, PV damage and water ingress. Implementing a covered carpark will also eliminate the potential for site user's car damage.
Extreme Temperature & Heatwaves	Rockhampton Aero (BOM Station 39083) is the closest weather station to the project (12-13km from project site). Numerous extreme weather events have been recorded such as: — Highest Recorded Temperature: 45.3ºC on 18 November 1990. — Recent Heatwaves: 42.3ºC on 13 February 2017 and 41.6ºC on 21 December 2019.	Temperatures exceeding 40ºC in recent years shows the project is expected to contribute to the urban heat island effect. The design should consider high albedo and cooling materials to reduce the effect of extreme temperatures to the CQU users.
Bushfire, Air Quality, and Drought	*Nov 2019*: Central QLD fires (e.g. Cobraball/Yeppoon) destroyed homes, and smoke severely affected the Rockhampton region.	This recent bushfire event and extreme temperatures anticipated in the Rockhampton region show the designs should consider potential smoke infiltration reduction strategies. Ember attack may also arise, with design features required to reduce the risk of building destruction.

4 Green Star Buildings

The Green Star certification is an established industry framework to develop, verify and assure sustainability in buildings, interiors and communities in Australia. The Green Star rating is determined by comparing the number of available points achieved out of the total available points. The rating scale shown below details the percentage thresholds for the Star ratings awarded.

As per the CQU general design requirements, buildings undergoing significant refurbishment (Buildings 33 and 36) are to meet “design requirements” for a minimum 4-star green star rating, with formal certification not required.

The Trades Building is deemed a ‘New Facility’ and therefore has a target to achieve an equivalent 5-star Green Star Buildings rating which demonstrates ‘Australian Excellence’ by being a high environmental performer that addresses social issues relevant to the building owner. The rating aims to meet current and future demands on the built environment with aspirational benchmarks for design, construction, and operational performance. It also considers the impacts against the key megatrends of the next decade, and to ensure it responds to the strategic goals of governments, tenants, investors, developers and building owners.

The design approach follows the environmental categories set out in the Green Star Buildings rating tool, including Responsible, Healthy, Resilient, Positive, Places, People, Nature, and Leadership which address sustainability holistically through design and construction.

The following section outlines potential Green Star criteria where additional points can be targeted to achieve the 35 credit points needed for a 5 Star Rating and 15 credit points for a 4 Star Rating across each of the Green Star Categories. Pathways for the new trades building and refurbishments are further detailed in Appendix A and Appendix B respectively

4.1 Minimum expectations

Based on the Green Star Submission Guidelines, all projects must meet a set of minimum expectations as a prerequisite to achieve any Green Star rating. For this project, both stages (new trade building and existing building refurbishments) must meet these minimum criteria, summarised in Table 4.1 to achieve either a 4 or 5 Star Green Star equivalent rating.

Table 4.1 Green Star Minimum Expectation Requirements

Credit	Criteria
Responsible construction	The builder must have an environmental management system, divert at least 80% of construction and demolition waste from landfill, and provide sustainability training to workers.
Verification & Handover	The building must include metering and monitoring systems, have set environmental targets, be tested for airtightness and be properly commissioned. Operations and maintenance information to be provided to facility managers and users.
Responsible Resource Management (operational waste)	The project team must demonstrate the building is designed to allow effective management of operational waste. At least three waste streams must be allowed for.
Clean air (ventilation system attributes)	The project is to provide high indoor air quality by separating and removing pollutants, cleaning ductwork, ensuring easy access for maintenance and supplying 50% more outdoor air than the standard requirement as required by AS 1668.2:2012.
Light quality/comfort	Lighting must be flicker-free, provide good colour rendering (CRI ≥ 85, R9 ≥ 50), and meet best-practice brightness and uniformity standards. Glare is to be reduced by shielding light sources from view. Spaces are designed for good daylight access and view, with a documented strategy for daylight, views and glare control.
Acoustic Comfort	Prepare an Acoustic Comfort Strategy describing how acoustic design delivers occupant comfort.
Exposure to toxins (Low VOC & Low Formaldehyde)	Paints, adhesives, sealants, carpets, and engineered wood products are low/non-toxic; occupants are not exposed to banned/highly toxic materials.
Climate change pre-screening	Complete the climate change pre-screening checklist and communicate the building’s exposure to climate risks.
Upfront Carbon Emissions	Achieves 10% less upfront carbon than a standard building.
Energy Use	At least 10% lower energy use than one built to NCC 2022
Zero Carbon Action Plan	Develop a Zero Carbon Action Plan signed off by the owner/developer and include in building operational documents.
Water Use	15% reduction in potable water compared to reference building (or minimum fittings performance).

Movement and place	Provide accessible, inclusive showers and changing facilities in a safe/protected space.
Inclusive Construction Practices	Provide gender-inclusive facilities/PPE and implement site policies to prevent discrimination, racism and bullying. The building is to be designed to be safe, inclusive, and accessible for everyone, with equitable access, clear wayfinding, and universally accessible amenities.
Impacts to nature	Building not built on or significantly impacting a site with high ecological value. Provide a Wetland Management Plan (if within/adjacent ≤100 m to wetland).
Light pollution	Complies with AS 4282:1997, Minimises upward light pollution; manage site wetland ecosystems;

4.2 Additional Green Star Criteria

To obtain a 4 or 5-star rating, a minimum of 15 or 35 points must be achieved respectively. The following section provides an overview of the potential Green Star criteria where additional points can be targeted to achieve a star rating equivalence.

The currently targeted points are part of an ongoing process, which will be revised further by the design team as the project is further developed to decide on the best mix of initiatives to achieve the required points. The exact points targeted are expected to change over the life of the project as the design develops and more information is known.

4.3 Responsible

The *Responsible* category recognises activities that ensure the building is designed, procured, built, and handed over in a responsible manner. Below are some initiatives the development is targeting:

- Engaging a Green Star Accredited Professional to ensure effective sustainability outcomes under Green Star certification.
- Target 90% of construction and demolition waste to be diverted from landfill.
- An independent commissioning agent (ICA) will be appointed to advice, monitor, and verify the design, planning, commissioning, and tuning activities to ensure the building systems are performing at their optimum. A soft landings approach will also be followed.
- Ensure at least 50% of structural components (by cost) meet a Responsible Products Value (RPV) of at least 10.
- Ensure that at least 40% of all internal building materials (finishes) by area meet a high level of sustainability by achieving a relevant sustainability attributes score of 7.

4.4 Healthy

The *Health* category recognises the improvement to the indoor environment quality to enhance the health and wellbeing of occupants. Below are some initiatives the development is targeting:

- Internal noise levels from services and the outside are limited, adequate acoustic separation and impact noise between floors is reduced are all achieved through an acoustic comfort strategy.

- To reduce building occupants being directly exposed to toxins the building will specify the use of low VOC paints, adhesives, sealants and carpets, and low formaldehyde engineered wood products within occupied spaces. On-site testing must be carried out to meet limits (e.g. TVOC and formaldehyde thresholds)

4.5 Resilient

The *Resilient* category enables risks that threaten the short- and long-term performance of the building have been considered through the design of the project. The development will demonstrate this by completing the following:

- Comprehensive, project-specific climate change risk and adaptation assessment will be prepared, ensuring that all high and extreme risks are identified and eliminated through design measures.
- At least 75% of the whole site area compromises of one or a combination of strategies that reduce the heat island effect, including light coloured roofing and extensive landscaping.

4.6 Positive

The *Positive* category acknowledges the value in understanding the full life cycle impacts of the building, which, in turn, can lead to better designs and material selection. Below are some initiatives the development is targeting:

- 10% less upfront carbon emissions compared to a standard building from materials and demolition works are offset.
- At least 20% lower energy use than one built under the NCC 2022 version reference.
- Eliminate or offset emissions from refrigerants.
- Achieve a 45% reduction in potable water usage when compared to a reference building via the inclusion of rainwater capture and reuse.
- 100% of the energy under the control of the building owner/operator, and all non-electricity energy provided for users that are not under the building owner's control, must be sourced from renewable.

4.7 Places

The *Places* category has an increased focus on putting people at the forefront of design. It focuses on the integration of the building into the urban fabric and delivers places that increase social cohesion. Below are some initiatives the development is targeting:

- Develop an urban context report and public realm interface design that outlines the urban context of the development and the design response to show the building's design contributes to the livability of the wider urban context and enhances the public realm.
- The building's design must reflect and celebrate local demographics and identities, the history of the place, and any hidden or minority entities via meaningful community engagement.

4.8 People

The *People* category encourages solutions that address the social health of the community by bringing a new dimension to the design and construction of buildings. Below are some initiatives the development is targeting:

- Developing and implementing construction practices to promote diversity and reduce physical and mental health impacts.

- The buildings design and construction must honor aboriginal and Torres Strait Islander culture by contributing to the organisations Reconciliation Action Plan or Incorporating Indigenous Design and Planning principles.
- At least 2% of the project’s capital expenditure will be allocated to procuring goods, services, and construction from Aboriginal businesses, social enterprises, and/or disability enterprises.

4.9 Nature

Worldwide and within Australia, rapid urbanisation is putting pressure on ecosystems and threatening biodiversity. The *Nature* category addresses this by rewarding designs that positively impact people and urban space. Below are some initiatives the development is targeting:

- At least 15% of the site must be landscaped with over 60% of indigenous plants and at least one significant tree per 500 sqm of landscape. A Biodiversity Management Plan is also required.

5 Conclusion

This ESD Report sets out how the proposed new Trades Building development and refurbishment of Building 33 and 36 at the CQU Rockhampton Tafe campus has considered sustainable design principles consistent with CQU's 2024-2028 Sustainability Framework and broader decarbonsiation goals.

This has been achieved through the holistic approach to sustainable design detailed in this report, incorporating passive design, efficient all-electric systems, low-carbon materials, improved indoor environmental quality, water-efficient fixtures and climate resilient design responses.

The new Trades Building is targeting a 5 Star Green Star Buildings equivalence, while the refurbishment works are aiming for a 4 Star Green Star Performance in line with the CQU's requirements for new and fully refurbished facilities. These targets are supported by compliance with NCC 2022 Section J, Queensland Development Code – Sustainable Buildings (QDC) and Rockhampton Regional Council Sustainability Strategy and Policies.

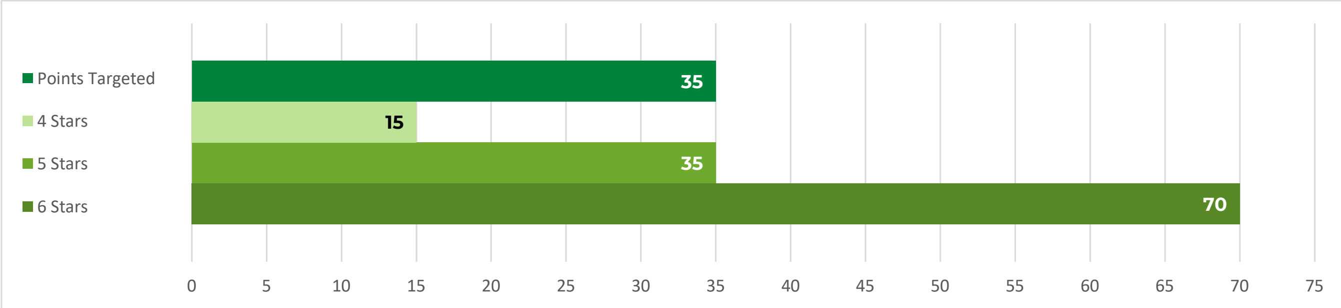
As the design progresses, the initiatives identified within this report will be further refined and coordinated across disciplines to ensure they remain appropriate, cost-effective, and aligned with CQU's long-term sustainability commitments.

Appendix A

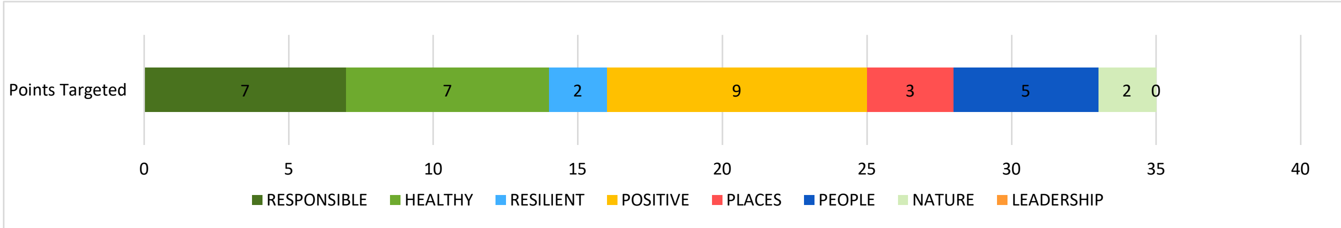
New Trades Building Green Star Pathway

Project	Rockhampton Tafe Trades
Stars	5 Star Pathway
Rev	3
Date	2/02/2026

Star Rating	Score
4 Stars	15
5 Stars	35
6 Stars	70
Points Targeted	35



Category	Total Points Available	Points Targeted
RESPONSIBLE	19	7
HEALTHY	12	7
RESILIENT	8	2
POSITIVE	35	9
PLACES	8	3
PEOPLE	9	5
NATURE	14	2
LEADERSHIP	10	0
Total	115	35



Risk	Points	Cumul. Points
Low	6	6
Medium	26	32
High	3	35

CA	Credit Achievement
ME	Minimum Expectation
EP	Exceptional Performance

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
RESPONSIBLE	1	Industry development	CA	Green Star Accredited Professional Financial Transparency Marketing Excellence	Engage a Green Star Accredited Professional contractually engaged as part of the core project team for the duration of the project and disclosure project cost and marketing information to the GBCA.	1	1	Low
			SS - CA	Collaborative Leasing	A high-quality green lease has been developed in line with the Better Buildings Partnership Leasing Standard, and opportunities for tenant collaboration are available.	1	0	Low
			SS -EP	Collaborative Leasing	A high-quality green lease has been developed in line with the Better Buildings Partnership Leasing Standard, and opportunities for tenant collaboration are available. Increased Tenant take up	1	0	Low
	2	Responsible construction	ME	Construction Management: EMS, EMP	The builder or head contractor has an environmental management system in place to manage its environmental impacts on site; The builder diverts at least 80% of construction and demolition waste from landfill; and The head contractor provides training on the sustainability targets of the building.	Minimum Expectation	Complies	Low
			CA	Construction and demolition waste	Diverting at least 90% of its construction and demolition waste from landfill. Compliance with Waste Reporting Criteria.	1	1	Low
	3	Verification & Handover	ME	Metering & Monitoring Commissioning and tuning Air Tightness and Building Information	The building is set up for optimum ongoing management due to its appropriate metering and monitoring systems. The building has set environmental performance targets, designed and tested for airtightness, been commissioned, and will be tuned. The project team create and deliver operations and maintenance information to the facilities management team at the time of handover. Information is available to building users on how to best use the building.	Minimum Expectation	Complies	High
			CA	Soft Landings and/or Independent Commissioning Agent	An independent level of verification is provided to the commissioning and tuning activities through the involvement of an independent commissioning agent, or through a soft landings approach that involves the future facilities management team. For large projects with a total building services value over $20M, both must occur.	1	1	Medium
	4	Responsible Resource Management	ME	Operational Waste	The project team must demonstrate the building is designed to allow effective management of operational waste. At least three waste streams must be allowed for	Minimum Expectation	Complies	Low
	5	Responsible procurement	CA	Procurement plan/strategy	Procurement process must follow ISO 20400 Guidelines and the project must demonstrate active efforts to address at least one identified risk and opportunity in its supply chain	1	0	
	6	Responsible structure	CA	Procurement of products	50% of all structural components (by cost) meet a Responsible Products Value of at least 10	3	3	High
			EP	Procurement of products	10% of all products in the structure (by cost) meet a Responsible Products Value of at least 15 or 80% of all products in the structure (by cost) meet a Responsible Products Value of at least 10	2	0	
	7	Responsible envelope	CA	Procurement of products	30% of all components in the building envelope (by cost) meet a Responsible Products Value of at least 10	2	0	
			EP	Procurement of products	10% of all products in the envelope (by cost) meet a Responsible Products Value of at least 15 or 60% of all products in the envelope (by cost) meet a Responsible Products Value of at least 10	2	0	
	8	Responsible systems	CA	Procurement of products	20% of all mechanical, hydraulic, transportation, and electrical systems (by cost) meet a Responsible Products Value of at least 6	1	0	
			EP	Procurement of products	5% of all active building systems (by cost) meet a Responsible Products Value of at least 11 or 35% of all active building systems (by cost) meet a Responsible Products Value of at least 6	1	0	

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
	9	Responsible finishes	CA	Procurement of products	40% of all internal building finishes by area must meet the relevant sustainability attributes score of 7	1	1	Medium
			EP	Procurement of products	10% of all internal building finishes (by cost) meet a Responsible Products Value of at least 12 or 60% of all internal building finishes (by cost) meet a Responsible Products Value of at least 7	1	0	
					TOTAL for Responsible	19	7	
HEALTHY	10	Clean air	ME	Ventilation System Attributes	Separation of Pollutants, Cleaning of Ductwork, Adequate Access to Both Sides of Moisture and Debris Catching Components and 50% improvement in provision of outdoor air as required by AS 1668:2:2012 to each space in the regularly occupied areas	Minimum Expectation	Complies	Medium
						Minimum Expectation	Complies	Medium
				Provision of outdoor air		Minimum Expectation	Complies	Medium
			CA	Ventilation System Attributes	Any mechanical ventilation system within the building, whether existing or new, must provide adequate access to both sides of all moisture and debris-catching components for maintenance within the air distribution system.	2	0	
				Provision of outdoor air	Outdoor air is provided at a rate 100% greater than the minimum required by AS 1668.2:2012, or CO2 concentrations are maintained below 700ppm The naturally ventilated space must meet the requirements of AS 1668.4:2012 under all likely weather conditions. Projects must justify how the nominated area will perform as a naturally ventilated space in these conditions.			
	11	Light quality	ME	Lighting Comfort	Lighting within the building must meet the following criteria: •All lighting must be flicker-free; •Light sources must have a minimum Colour Rendering Index (CRI) average R1 to R8 of 85 or higher, and have a CRI R9 of 50 or higher; •Light sources must meet best practice illuminance levels for each task within each space type with a maintained illuminance that meets the levels recommended in AS/NZS 1680.1:2006 series applicable to the project type and including maintenance; •The maintained Illuminance values must achieve a uniformity of no less than that specified in Table 3.2 of AS/NZS 1680.1:2006, with a maintenance factor method as defined in AS/NZS 1680.4.; and •All light sources must have a minimum of 3 MacAdam Ellipses.	Minimum Expectation	Complies	Medium
				Glare	Bare light sources must be fitted with baffles, louvers, translucent diffusers, ceiling design, or other means that obscures the direct light source from all viewing angles of occupants, including occupants looking directly upwards.	Minimum Expectation	Complies	Medium
				Daylight	Regularly occupied spaces are to within a reasonable proximity to glazed facades, windows or skylights. Building occupants are to be provided with unrestricted access to daylit indoor common spaces. A narrative describing the building's daylight, view and glare control strategy;	Minimum Expectation	Complies	Medium
			CA	Daylight	For non-residential buildings, at least 40% of the principle averaged across the building must receive high levels of daylight with no less than 20% on any floor or tenancy (whichever is smaller). For residential buildings, 60% of the combined living and bedroom area of each apartment unit must comply with the daylight requirements. Kitchens are not included in the calculations. The daylight levels must also be present in at least 20% of the area of each bedroom and living area. Residential buildings and hospitality buildings must provide room blackout blinds or curtains to all bedrooms. If blinds or curtains are part of a packaged décor, all blinds offered for the bedroom décor must be blackout blinds.	2	2	Medium
				Artificial lighting	The artificial lighting solution must address the quality of light in the space, provide highlights and contrast, and seek to avoid excessive lighting or overly uniform solutions. Compliance should be demonstrated across 95% of the nominated area. •The walls within the field of view of occupants in regularly occupied spaces must have an average surface reflectance value of 0.70 and an average surface illuminance of at least 50% of the horizontal illuminance levels required for task, and; •Vertical illuminance in workspaces: ensure that 50% of the horizontal task illuminance reaches the average eye height for 90% of primary spaces using vertical illuminance calculation grid. The illuminance values must be calculated in accordance with AS/NZS 1680 series for the relevant task. Where unknown, a conservative estimate can be used.	0	0	
			EP	Daylight and artificial lighting	Achieves both Credit achievements criteria	0	0	
	12	Acoustic Comfort	ME	Acoustic Comfort	An Acoustic Comfort Strategy is prepared to describe how the building and acoustic design aims to deliver acoustic comfort to the building occupants.	Minimum Expectation	Complies	Low
			CA	Internal Noise	Internal ambient noise levels in the nominated areas must be no less than 5 dB below the lower range value and no greater than the upper range value relevant to the activity type in each space as recommended in AS/NZS 2107	2	2	Medium
					In residential dwellings the internal ambient noise levels can exclude those services under the direct control of the occupant such as air-conditioning units and switchable exhaust fans (e.g. toilet, kitchen hoods and laundries). In buildings with sleeping areas (e.g. residential, hotel, hospitals, etc), to achieve the Internal Noise performance requirements of this credit, noise levels must not exceed recommended Sleep Disturbance criteria as defined in the NSW EPA Road Noise Policy 2011: Up to two noise events per night: maximum internal noise levels below 70 dB LAmax; and All other events: maximum internal noise levels below 55 dB LAmax			
					The project must address noise transmission between enclosed spaces within the nominated area.			
					The sound insulation between internal spaces complies with: Dw + LAeqT > X.			

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
				Acoustic Separation	All walls and floors (excluding riser walls) separating enclosed spaces must exceed the minimum NCC requirements by 5points (excluding impact noise) Party walls separating open plan kitchens (where joinery units are fixed) from another open plan kitchen/living room shall be discontinuous in construction (discontinuous in accordance with the National Construction Code); and Entry doors must have perimeter and threshold seals.			
				Impact Noise Transfer	Impact noise transfer measured in accordance with ISO 16283-2 through a floor where: Floors are located above nominated areas; or Adjacent spaces belonging to different tenancies which share a floor must not exceed dB LnT,w: - 55 for floors above residential accommodation spaces and - 60 for all other spaces			
	13	Exposure to toxins	ME	Low VOC Products, Formaldehyde - Engineered Wood Products	The building's paints adhesives, sealants, carpets, and engineered wood products are low or non-toxic. Occupants are not exposed to banned or highly toxic materials in the building.	Minimum Expectation	Complies	Medium
			CA	VOC level testing	Onsite test meeting following limits: TVOC = 0.27ppm; Formaldehyde = 0.02ppm. 6 test locations, 3 per floor	2	2	Medium
	14	Amenity and comfort	CA	Amenities	Provide a quiet room, and/or exercise room and/or a parents room (with support for lactation activities) for occupants at rate of 1m² for every 10 people with individual rooms no smaller than 10m²	2	0	
	15	Human connection to nature	CA	Views, plants, nature design	Views, Nature inspired design/plant OR Interaction with nature	1	1	Medium
			EP	Allocation to nature	5% of the buildings floor area or site area (whichever is greater) is utilised for connection to nature including green walls and roof gardens.	1	0	
					TOTAL for Healthy	**12**	**7**	
RESILIENCE	16	Climate change resilience	ME	Climate Change Pre-screening	The project team completes the climate change pre-screening checklist. The project team communicates the building's exposure to climate change risks to the applicant.	Minimum Expectation	Complies	Low
			CA	Comprehensive climate assessment	Comprehensive project-specific climate change risk and adaptation assessment must be developed for the project. High and extreme risks designed out.	1	1	Medium
	17	Operations resilience	CA	Broad resilience assessment	The building's design and future operational plan addresses any high or extreme system-level interdependency risks. The building's design maintains a level of survivability and design purpose in a blackout.	2	0	
	18	Community resilience	CA	Community needs assessment	Review of key vulnerabilities within the community it is located and take steps to build community resilience through external consultation	1	0	
	19	Heat Resilience	CA	Heat Resilience	At least 75% of the whole site area comprises of one or a combination of strategies that reduce the heat island effect.	1	1	Low
	20	Grid Resilience	CA	Peak energy demand	The building is designed to reduced maximum demand by 10% with • Provides active generation and storage systems and the infrastructure to deliver an appropriate demand response strategy. The system (generation or storage) must incorporate switch gear and transfer switches to enable it to operate in the event of grid outage or grid demand response event; or • Has reduced its electricity consumption through passive design.	3	0	
					TOTAL for Resilient	**8**	**2**	
	21	Upfront Carbon Emissions	ME	Improvement over reference	Emits 10% less upfront carbon emissions compared to a standard building	Minimum Expectation	Complies	Medium
			CA		Emits 20% less upfront carbon emissions compared to a standard building Demolition works are offset	3	0	
			EP		Emits 40% less upfront carbon emissions compared to a standard building Demolition works are offset	3	0	
	22	Energy use	ME	Improvement over reference	At least a 10% lower energy use than one built to the National Construction Code 2019.	Minimum Expectation	Complies	Low
			CA		At least a 20% lower energy use than one built to the National Construction Code 2019.	3	3	Medium
			EP		At least a 30% lower energy use than one built to the National Construction Code 2019.	3	0	
			ME		The project team must develop a Zero Carbon Action Plan for the building. The plan must be signed off by the building owner or developer and included in any operational documents for the building.	Minimum Expectation	Complies	Medium

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
POSITIVE	23	Energy Source	CA	Zero Carbon Action Plan	100% of the building's electricity must come from renewables. On-site or off-site renewables are acceptable.	3	3	Medium
			EP		All energy consumed by the building comes from renewables. Energy is defined as all electricity consumed, as well as any fuels burned on-site, or off-site, for power, heating, cooling and cooking	3	0	
			SS - CA		Sector Specific Credit Achievment - Tenant Energy Source - The building owner actively assist tenants to procure renewable electricity - At least 40% of tenant space (by NLA) uses 100% renewable electricity	2	0	
			SS - EP		Sector Specific Exceptional Performance - Tenant Energy Source - At least 80% of tenant space (by NLA) uses 100% renewable energy	3	0	
	24	Other Carbon Emissions	CA	No refrigerants or offsets	Eliminates or offsets emissions from refrigerants. Emissions are assumed to be the refrigerant charge multiplied by its Global Warming Potential. The emissions must be offset at 100%. This includes fridges/freezers provided as part of a residential fitout package.	2	0	
			EP	Other carbon sources not captured	The building owner eliminates or offsets additional emissions not captured in the rest of the Positive category.	2	0	
	25	Water use	ME	Water use	15% reduction in potable water usage when compared to a reference building, or minimum fittings performance	Minimum Expectation	Complies	Low
						Minimum Expectation	Complies	Low
			CA		45% reduction in potable water usage when compared to a reference building,	3	3	Low
			EP		75% reduction in potable water usage when compared to a reference building,	3	0	
	26	Life Cycle Impacts	CA	Life Cycle Assessment	Comparative life cycle assessment (LCA) must be conducted for the project. The project demonstrates a 30% reduction in life cycle impacts when compared to standard practice.	2	0	
					TOTAL for Positive	**35**	**9**	
PLACES	27	Movement and Place	ME	End of Trip facilities	The building includes showers and changing facilities for building occupants that are accessible, inclusive and located in a safe and protected space.	Minimum Expectation	Complies	Medium
			CA	Cyclist Facilities	The building's access must prioritise walking and cycling options. This means the building's access must be well lit, weather protected, and separated from vehicles. The building must also include access to cyclist facilities that are separated from the primary vehicle entrance to ensure safety.	3	0	
				Sustainable Transport	Project must prepare and implement a Sustainable Transport Plan and be reflected in building's design. The building's design and location prioritises walking, cycling, and transport options that reduce the need for private fossil fuel powered vehicles.			
				EV charging provision	Ready to charge EV charging points to at least 5% of all car parking spaces.			
				Reducing Private Vehicle Use	Using the inputs from the Sustainable Transport Plan to complete the GBCA's Movement and Place calculator, the building's design and location must achieve: - Emission reduction: 40% - Active mode encouragement: 90% - Vehicle Kilometres Travelled (VKT) reduction: 20%			
				Encouraging Walkability	The building's design and location must encourage walking to and from a number of amenities. This means designing roads within the site boundary to prioritise pedestrians, and either providing within, or being located close to, a number of amenities.			
	28	Enjoyable places	CA	Provision of communal spaces	The building delivers memorable, beautiful, vibrant communal or public places where people want to gather and participate in the community. The spaces are inclusive, safe, flexible and enjoyable. 0.25 m2/ occupant or 2.5% of GFA, whichever is greater shall be dedicated for this purpose.	2	0	
	29	Contribution to place	CA	Urban Context Report	The building's design contributes to the liveability of the wider urban context and enhances the public realm demonstrated by an urban context analysis or independent design review.	2	2	Medium
	30	Culture, Heritage and Identity	CA	Culture, Heritage and Identity	The building's design reflects and celebrates local demographics and identities, the history of the place, and any hidden or minority entities. This celebration was arrived through meaningful engagement with community groups early in the design process.	1	1	Medium
					TOTAL for Places	**8**	**3**	
PEOPLE	31	Inclusive Construction Practices	ME	High quality staff support	During the building's construction, the head contractor provides gender inclusive facilities and protective equipment. The head contractor also installs policies on-site to increase awareness and reduces instances of discrimination, racism and bullying.	Minimum Expectation	Complies	Low
			CA		Promote positive mental and physical health outcomes of the site activities and culture of site workers, through programs and solutions on site that address at least 5 areas	1	1	Medium
	32	Indigenous Inclusion	CA	Indigenous Inclusion	The building's design and construction celebrates Aboriginal and Torres Strait Islander people, culture and heritage by undertaking one or both of the following: Playing an active role in the organisational Reconciliation Action Plan; or Incorporating design elements using the Indigenous Design and Planning principle	2	2	Medium
	33	Procurement and Workforce Inclusion	CA	Procurement of goods and services	Direct at least 2% of project's CAPEX to the procurement of goods, services and construction provided by Aboriginal businesses; Social enterprises; and/ or Disability enterprises	2	2	Medium
			EP	Procurement of goods and services	Direct at least 4% of project's CAPEX to the procurement of goods, services and construction provided by Aboriginal businesses; Social enterprises; and/ or Disability enterprises	1	0	
	34	Design for inclusion	CA	Designing for inclusion	Show an increase in design for diversity considerations beyond legislative requirements in terms of equal access, diverse wayfinding and inclusive spaces; training also required	2	0	
			EP	Community engagement	project team must consult with distinct community types to develop a needs analysis that will influence the project.	1	0	
					TOTAL for People	**9**	**5**	

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
NATURE	35	Impacts to nature	ME	Impacts to nature	The building was not built on, or significantly impacted, a site with a high ecoligical value	Minimum Expectation	Complies	Low
				Light Pollution to Night Sky	The building was not built on, or significantly impacted, a site with a high ecological value. Light pollution minimised. Management of the sites wetland ecosystems.	Minimum Expectation	Complies	Low
				Wetland Management Plan	If the project sit is adjacent, within 100 meters, or contains a wetland	Minimum Expectation	Complies	Low
			CA	Ecological report	The building's design and construction conserves existing natural soil, hydrological flows, and vegetation elements. If deemed necessary (area of high biodiversity value) by an Ecologist, at least 50% of the existing site with high biodiversity value is retained.	2	0	
	36	Biodiversity enhancement	CA	Landscape selection and provision	External landscape at a ratio of either 15% of the site area or at a ratio of 1:500 of GFA, whichever is larger. Greater than 60% of plants must be indigenous, and include 1 significant tree per 500m2 landscape. A Biodiversity Management Plan must be developed.	2	2	Medium
			EP		External landscape at a ratio of either 30% of the site area or at a ratio of 1:300 of GFA, whichever is larger. Greater than 80% of plants must be indigenous. Tree beer 250m2 site.	2	0	
	37	Nature connectivity	CA	Species Connectivity	Provide either landscaping or infrastructure to promote movement. Each conservation area must be at least 185m2. Connect to green/blue grid strategy where relevant for the site.	2	0	
	38	Nature Stewardship	CA	Nature Stewardship	Restore offsite biodiversity equivalent to the total GFA of the development, or site area, whichever is greater. Offsite area must be same ecological value.	2	0	
	39	Waterway protection	CA	Waterway protection	Demonstrate an annual average flow reduction (ML/yr) of 40% compared to pre-development levels and meets specified pollution targets.	2	0	
			EP	Waterway protection	Demonstrate an annual average flow reduction (ML/yr) of 80% compared to pre-development levels and meets specified pollution targets.	2	0	
					TOTAL for Nature	**14**	**2**	
LEADERSHIP	40	Market Transformation	CA	Market Transformation	The project demonstrates: How a building solution or process is considered leading in their targeted sector, nationally or globally; or That the technology or process is not commonly used within Australia's building industry; or globally, depending on the context of the innovation claimed.	5	0	
	41	Leadership Challenges	CA	Leadership Challenges	Meet the requirements of the following innovation challenges identified by the GBCA	5	0	
					TOTAL for Leadership	**10**	**0**	

Appendix B

Building 33 and 36 Refurbishment Green Star Pathway

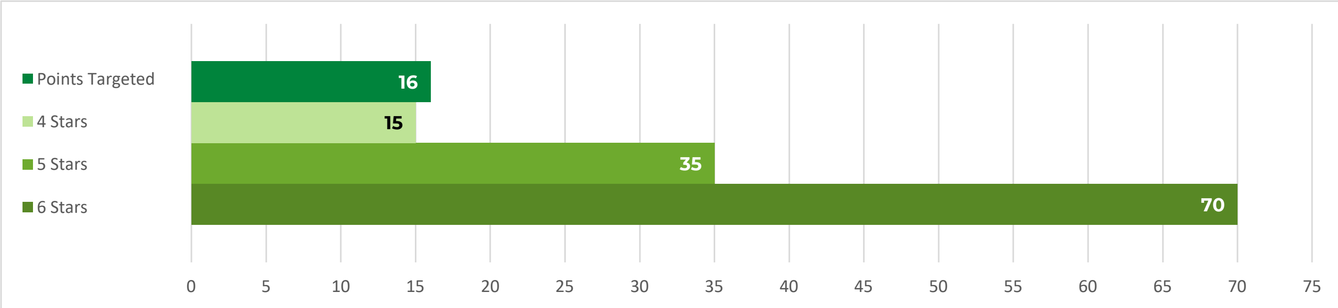
Project	Rockhampton Tafe Building 33 and 36 Refurbushment
Stars	4 Star Pathway
Rev	2
Date	2/02/2026

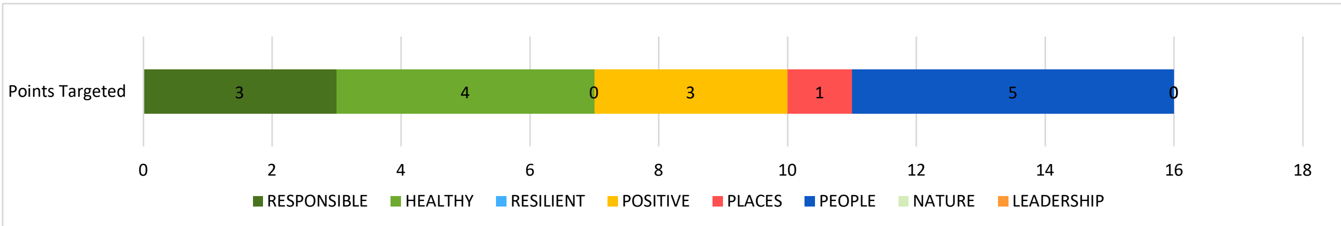
Star Rating	Score
4 Stars	15
5 Stars	35
6 Stars	70
Points Targeted	16

Category	Total Points Available	Points Targeted
RESPONSIBLE	19	3
HEALTHY	11	4
RESILIENT	8	0
POSITIVE	35	3
PLACES	8	1
PEOPLE	9	5
NATURE	14	0
LEADERSHIP	10	0
Total	114	16

Risk	Points	Cumul. Points
Low	2	2
Medium	14	16
High	3	19

CA	Credit Achievement
ME	Minimum Expectation
EP	Exceptional Performance





Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
RESPONSIBLE	1	Industry development	CA	Green Star Accredited Professional Financial Transparency Marketing Excellence	Engage a Green Star Accredited Professional contractually engaged as part of the core project team for the duration of the project and disclosure project cost and marketing information to the GBCA.	1	1	Low
			SS - CA	Collaborative Leasing	A high-quality green lease has been developed in line with the Better Buildings Partnership Leasing Standard, and opportunities for tenant collaboration are available.	1	0	Low
			SS -EP	Collaborative Leasing	A high-quality green lease has been developed in line with the Better Buildings Partnership Leasing Standard, and opportunities for tenant collaboration are available. Increased Tenant take up	1	0	Low
	2	Responsible construction	ME	Construction Management: EMS, EMP	The builder or head contractor has an environmental management system in place to manage its environmental impacts on site; The builder diverts at least 80% of construction and demolition waste from landfill; and The head contractor provides training on the sustainability targets of the building.	Minimum Expectation	Complies	Low
			CA	Construction and demolition waste	Diverting at least 90% of its construction and demolition waste from landfill. Compliance with Waste Reporting Criteria.	1	1	Low
	3	Verification & Handover	ME	Metering & Monitoring Commissioning and tuning Air Tightness and Building Information	The building is set up for optimum ongoing management due to its appropriate metering and monitoring systems. The building has set environmental performance targets, designed and tested for airtightness, been commissioned, and will be tuned. The project team create and deliver operations and maintenance information to the facilities management team at the time of handover. Information is available to building users on how to best use the building.	Minimum Expectation	Complies	High
			CA	Soft Landings and/or Independent Commissioning Agent	An independent level of verification is provided to the commissioning and tuning activities through the involvement of an independent commissioning agent, or through a soft landings approach that involves the future facilities management team. For large projects with a total building services value over $20M, both must occur.	1	0	Medium
	4	Responsible Resource Management	ME	Operational Waste	The project team must demonstrate the building is designed to allow effective management of operational waste. At least three waste streams must be allowed for	Minimum Expectation	Complies	Low
	5	Responsible procurement	CA	Procurement plan/strategy	Procurement process must follow ISO 20400 Guidelines and the project must demonstrate active efforts to address at least one identified risk and opportunity in its supply chain	1	0	
	6	Responsible structure	CA	Procurement of products	50% of all structural components (by cost) meet a Responsible Products Value of at least 10	3	0	
			EP	Procurement of products	10% of all products in the structure (by cost) meet a Responsible Products Value of at least 15 or 80% of all products in the structure (by cost) meet a Responsible Products Value of at least 10	2	0	
	7	Responsible envelope	CA	Procurement of products	30% of all components in the building envelope (by cost) meet a Responsible Products Value of at least 10	2	0	
			EP	Procurement of products	10% of all products in the envelope (by cost) meet a Responsible Products Value of at least 15 or 60% of all products in the envelope (by cost) meet a Responsible Products Value of at least 10	2	0	
			CA	Procurement of products	20% of all mechanical, hydraulic, transportation, and electrical systems (by cost) meet a Responsible Products Value of at least 6	1	0	

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
	8	**Responsible systems**	EP	Procurement of products	5% of all active building systems (by cost) meet a Responsible Products Value of at least 11 or 35% of all active building systems (by cost) meet a Responsible Products Value of at least 6	1	0	
	9	**Responsible finishes**	CA	Procurement of products	40% of all internal building finishes by area must meet the relevant sustainability attributes score of 7	1	1	Medium
			EP	Procurement of products	10% of all internal building finishes (by cost) meet a Responsible Products Value of at least 12 or 60% of all internal building finishes (by cost) meet a Responsible Products Value of at least 7	1	0	
					TOTAL for Responsible	**19**	**3**	
HEALTHY	10	**Clean air**	ME	Ventilation System Attributes	Separation of Pollutants, Cleaning of Ductwork, Adequate Access to Both Sides of Moisture and Debris Catching Components and 50% improvement in provision of outdoor air as required by AS 1668:2:2012 to each space in the regularly occupied areas	Minimum Expectation	Complies	Medium
						Minimum Expectation	Complies	Medium
				Provision of outdoor air		Minimum Expectation	Complies	Medium
			CA	Ventilation System Attributes	Any mechanical ventilation system within the building, whether existing or new, must provide adequate access to both sides of all moisture and debris-catching components for maintenance within the air distribution system.	2	0	
				Provision of outdoor air	Outdoor air is provided at a rate 100% greater than the minimum required by AS 1668.2:2012, or CO2 concentrations are maintained below 700ppm The naturally ventilated space must meet the requirements of AS 1668.4:2012 under all likely weather conditions. Projects must justify how the nominated area will perform as a naturally ventilated space in these conditions.			
	11	**Light quality**	ME	Lighting Comfort	Lighting within the building must meet the following criteria: •All lighting must be flicker-free; •Light sources must have a minimum Colour Rendering Index (CRI) average R1 to R8 of 85 or higher, and have a CRI R9 of 50 or higher; •Light sources must meet best practice illuminance levels for each task within each space type with a maintained illuminance that meets the levels recommended in AS/NZS 1680.1:2006 series applicable to the project type and including maintenance; •The maintained Illuminance values must achieve a uniformity of no less than that specified in Table 3.2 of AS/NZS 1680.1:2006, with a maintenance factor method as defined in AS/NZS 1680.4.; and •All light sources must have a minimum of 3 MacAdam Ellipses.	Minimum Expectation	Complies	Medium
				Glare	Bare light sources must be fitted with baffles, louvers, translucent diffusers, ceiling design, or other means that obscures the direct light source from all viewing angles of occupants, including occupants looking directly upwards.	Minimum Expectation	Complies	Medium
				Daylight	Regularly occupied spaces are to within a reasonable proximity to glazed facades, windows or skylights. Building occupants are to be provided with unrestricted access to daylit indoor common spaces. A narrative describing the building’s daylight, view and glare control strategy;	Minimum Expectation	Complies	Medium
			CA	Daylight	For non-residential buildings, at least 40% of the principle averaged across the building must receive high levels of daylight with no less than 20% on any floor or tenancy (whichever is smaller). For residential buildings, 60% of the combined living and bedroom area of each apartment unit must comply with the daylight requirements. Kitchens are not included in the calculations. The daylight levels must also be present in at least 20% of the area of each bedroom and living area. Residential buildings and hospitality buildings must provide room blackout blinds or curtains to all bedrooms. If blinds or curtains are part of a packaged décor, all blinds offered for the bedroom décor must be blackout blinds.	1	0	
				Artificial lighting	The artificial lighting solution must address the quality of light in the space, provide highlights and contrast, and seek to avoid excessive lighting or overly uniform solutions. Compliance should be demonstrated across 95% of the nominated area. •The walls within the field of view of occupants in regularly occupied spaces must have an average surface reflectance value of 0.70 and an average surface illuminance of at least 50% of the horizontal illuminance levels required for task, and; •Vertical illuminance in workspaces: ensure that 50% of the horizontal task illuminance reaches the average eye height for 90% of primary spaces using vertical illuminance calculation grid. The illuminance values must be calculated in accordance with AS/NZS 1680 series for the relevant task. Where unknown, a conservative estimate can be used.	0	0	
			EP	Daylight and artificial lighting	Achieves both Credit achievements criteria	0	0	
	12	**Acoustic Comfort**	ME	Acoustic Comfort	An Acoustic Comfort Strategy is prepared to describe how the building and acoustic design aims to deliver acoustic comfort to the building occupants.	Minimum Expectation	Complies	Low
			CA	Internal Noise	Internal ambient noise levels in the nominated areas must be no less than 5 dB below the lower range value and no greater than the upper range value relevant to the activity type in each space as recommended in AS/NZS 2107	2	2	Medium
					In residential dwellings the internal ambient noise levels can exclude those services under the direct control of the occupant such as air-conditioning units and switchable exhaust fans (e.g. toilet, kitchen hoods and laundries). In buildings with sleeping areas (e.g. residential, hotel, hospitals, etc), to achieve the Internal Noise performance requirements of this credit, noise levels must not exceed recommended Sleep Disturbance criteria as defined in the NSW EPA Road Noise Policy 2011: Up to two noise events per night: maximum internal noise levels below 70 dB LAmax; and All other events: maximum internal noise levels below 55 dB LAmax			
					The project must address noise transmission between enclosed spaces within the nominated area.			
					The sound insulation between internal spaces complies with: Dw + LAeqT > X.			

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
				Acoustic Separation	All walls and floors (excluding riser walls) separating enclosed spaces must exceed the minimum NCC requirements by 5points (excluding impact noise) Party walls separating open plan kitchens (where joinery units are fixed) from another open plan kitchen/living room shall be discontinuous in construction (discontinuous in accordance with the National Construction Code); and Entry doors must have perimeter and threshold seals.			
				Impact Noise Transfer	Impact noise transfer measured in accordance with ISO 16283-2 through a floor where: Floors are located above nominated areas; or Adjacent spaces belonging to different tenancies which share a floor must not exceed dB LnT,w: - 55 for floors above residential accommodation spaces and - 60 for all other spaces			
	13	Exposure to toxins	ME	Low VOC Products, Formaldehyde - Engineered Wood Products	The building's paints adhesives, sealants, carpets, and engineered wood products are low or non-toxic. Occupants are not exposed to banned or highly toxic materials in the building.	Minimum Expectation	Complies	Medium
			CA	VOC level testing	Onsite test meeting following limits: TVOC = 0.27ppm; Formaldehyde = 0.02ppm. 6 test locations, 3 per floor	2	2	Medium
	14	Amenity and comfort	CA	Amenities	Provide a quiet room, and/or exercise room and/or a parents room (with support for lactation activities) for occupants at rate of 1m² for every 10 people with individual rooms no smaller than 10m²	2	0	
	15	Human connection to nature	CA	Views, plants, nature design	Views, Nature inspired design/plant OR Interaction with nature	1	0	Medium
			EP	Allocation to nature	5% of the buildings floor area or site area (whichever is greater) is utilised for connection to nature including green walls and roof gardens.	1	0	
					TOTAL for Healthy	**11**	**4**	
RESILIENCE	16	Climate change resilience	ME	Climate Change Pre-screening	The project team completes the climate change pre-screening checklist. The project team communicates the building's exposure to climate change risks to the applicant.	Minimum Expectation	Complies	Low
			CA	Comprehensive climate assessment	Comprehensive project-specific climate change risk and adaptation assessment must be developed for the project. High and extreme risks designed out.	1	0	
	17	Operations resilience	CA	Broad resilience assessment	The building's design and future operational plan addresses any high or extreme system-level interdependency risks. The building's design maintains a level of survivability and design purpose in a blackout.	2	0	
	18	Community resilience	CA	Community needs assessment	Review of key vulnerabilities within the community it is located and take steps to build community resilience through external consultation	1	0	
	19	Heat Resilience	CA	Heat Resilience	At least 75% of the whole site area comprises of one or a combination of strategies that reduce the heat island effect.	1	0	
	20	Grid Resilience	CA	Peak energy demand	The building is designed to reduced maximum demand by 10% with • Provides active generation and storage systems and the infrastructure to deliver an appropriate demand response strategy. The system (generation or storage) must incorporate switch gear and transfer switches to enable it to operate in the event of grid outage or grid demand response event; or • Has reduced its electricity consumption through passive design.	3	0	
					TOTAL for Resilient	**8**	**0**	
	21	Upfront Carbon Emissions	ME	Improvement over reference	Emits 10% less upfront carbon emissions compared to a standard building	Minimum Expectation	Complies	Medium
			CA		Emits 20% less upfront carbon emissions compared to a standard building Demolition works are offset	3	0	
			EP		Emits 40% less upfront carbon emissions compared to a standard building Demolition works are offset	3	0	
	22	Energy use	ME	Improvement over reference	At least a 10% lower energy use than one built to the National Construction Code 2019.	Minimum Expectation	Complies	Low
			CA		At least a 20% lower energy use than one built to the National Construction Code 2019.	3	0	
			EP		At least a 30% lower energy use than one built to the National Construction Code 2019.	3	0	
			ME		The project team must develop a Zero Carbon Action Plan for the building. The plan must be signed off by the building owner or developer and included in any operational documents for the building.	Minimum Expectation	Complies	Medium

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
POSITIVE	23	Energy Source	CA	Zero Carbon Action Plan	100% of the building's electricity must come from renewables. On-site or off-site renewables are acceptable.	3	3	Medium
			EP		All energy consumed by the building comes from renewables. Energy is defined as all electricity consumed, as well as any fuels burned on-site, or off-site, for power, heating, cooling and cooking	3	0	
			SS - CA		Sector Specific Credit Achievement - Tenant Energy Source - The building owner actively assist tenants to procure renewable electricity - At least 40% of tenant space (by NLA) uses 100% renewable electricity	2	0	
			SS - EP		Sector Specific Exceptional Performance - Tenant Energy Source - At least 80% of tenant space (by NLA) uses 100% renewable energy	3	0	
	24	Other Carbon Emissions	CA	No refrigerants or offsets	Eliminates or offsets emissions from refrigerants. Emissions are assumed to be the refrigerant charge multiplied by its Global Warming Potential. The emissions must be offset at 100%. This includes fridges/freezers provided as part of a residential fitout package.	2	0	
			EP	Other carbon sources not captured	The building owner eliminates or offsets additional emissions not captured in the rest of the Positive category.	2	0	
	25	Water use	ME	Water use	15% reduction in potable water usage when compared to a reference building, or minimum fittings performance	Minimum Expectation	Complies	Low
						Minimum Expectation	Complies	Low
			CA		45% reduction in potable water usage when compared to a reference building,	3	0	
			EP		75% reduction in potable water usage when compared to a reference building,	3	0	
	26	Life Cycle Impacts	CA	Life Cycle Assessment	Comparative life cycle assessment (LCA) must be conducted for the project. The project demonstrates a 30% reduction in life cycle impacts when compared to standard practice.	2	0	
					TOTAL for Positive	**35**	**3**	
PLACES	27	Movement and Place	ME	End of Trip facilities	The building includes showers and changing facilities for building occupants that are accessible, inclusive and located in a safe and protected space.	Minimum Expectation	Complies	Medium
			CA	Cyclist Facilities	The building's access must prioritise walking and cycling options. This means the building's access must be well lit, weather protected, and separated from vehicles. The building must also include access to cyclist facilities that are separated from the primary vehicle entrance to ensure safety.	3	0	
				Sustainable Transport	Project must prepare and implement a Sustainable Transport Plan and be reflected in building's design. The building's design and location prioritises walking, cycling, and transport options that reduce the need for private fossil fuel powered vehicles.			
				EV charging provision	Ready to charge EV charging points to at least 5% of all car parking spaces.			
				Reducing Private Vehicle Use	Using the inputs from the Sustainable Transport Plan to complete the GBCA's Movement and Place calculator, the building's design and location must achieve: - Emission reduction: 40% - Active mode encouragement: 90% - Vehicle Kilometres Travelled (VKT) reduction: 20%			
				Encouraging Walkability	The building's design and location must encourage walking to and from a number of amenities. This means designing roads within the site boundary to prioritise pedestrians, and either providing within, or being located close to, a number of amenities.			
	28	Enjoyable places	CA	Provision of communal spaces	The building delivers memorable, beautiful, vibrant communal or public places where people want to gather and participate in the community. The spaces are inclusive, safe, flexible and enjoyable. 0.25 m2/ occupant or 2.5% of GFA, whichever is greater shall be dedicated for this purpose.	2	0	
	29	Contribution to place	CA	Urban Context Report	The building's design contributes to the liveability of the wider urban context and enhances the public realm demonstrated by an urban context analysis or independent design review.	2	0	
	30	Culture, Heritage and Identity	CA	Culture, Heritage and Identity	The building's design reflects and celebrates local demographics and identities, the history of the place, and any hidden or minority entities. This celebration was arrived through meaningful engagement with community groups early in the design process.	1	1	Medium
					TOTAL for Places	**8**	**1**	
PEOPLE	31	Inclusive Construction Practices	ME	High quality staff support	During the building's construction, the head contractor provides gender inclusive facilities and protective equipment. The head contractor also installs policies on-site to increase awareness and reduces instances of discrimination, racism and bullying.	Minimum Expectation	Complies	Low
			CA		Promote positive mental and physical health outcomes of the site activities and culture of site workers, through programs and solutions on site that address at least 5 areas	1	1	Medium
	32	Indigenous Inclusion	CA	Indigenous Inclusion	The building's design and construction celebrates Aboriginal and Torres Strait Islander people, culture and heritage by undertaking one or both of the following: Playing an active role in the organisational Reconciliation Action Plan; or Incorporating design elements using the Indigenous Design and Planning principle	2	2	Medium
	33	Procurement and Workforce Inclusion	CA	Procurement of goods and services	Direct at least 2% of project's CAPEX to the procurement of goods, services and construction provided by Aboriginal businesses; Social enterprises; and/ or Disability enterprises	2	2	Medium
			EP	Procurement of goods and services	Direct at least 4% of project's CAPEX to the procurement of goods, services and construction provided by Aboriginal businesses; Social enterprises; and/ or Disability enterprises	1	0	
	34	Design for inclusion	CA	Designing for inclusion	Show an increase in design for diversity considerations beyond legislative requirements in terms of equal access, diverse wayfinding and inclusive spaces; training also required	2	0	
			EP	Community engagement	project team must consult with distinct community types to develop a needs analysis that will influence the project.	1	0	
					TOTAL for People	**9**	**5**	

Category	Number	Credit	Credit Type	Pathway	Criteria	Total Points Available	Points Targeted	Risk Rating
NATURE	35	Impacts to nature	ME	Impacts to nature	The building was not built on, or significantly impacted, a site with a high ecoligical value	Minimum Expectation	Complies	Low
				Light Pollution to Night Sky	The building was not built on, or significantly impacted, a site with a high ecological value. Light pollution minimised. Management of the sites wetland ecosystems.	Minimum Expectation	Complies	Low
				Wetland Management Plan	If the project sit is adjacent, within 100 meters, or contains a wetland	Minimum Expectation	Complies	Low
			CA	Ecological report	The building's design and construction conserves existing natural soil, hydrological flows, and vegetation elements. If deemed necessary (area of high biodiversity value) by an Ecologist, at least 50% of the existing site with high biodiversity value is retained.	2	0	
	36	Biodiversity enhancement	CA	Landscape selection and provision	External landscape at a ratio of either 15% of the site area or at a ratio of 1:500 of GFA, whichever is larger. Greater than 60% of plants must be indigenous, and include 1 significant tree per 500m2 landscape. A Biodiversity Management Plan must be developed.	2	0	
			EP		External landscape at a ratio of either 30% of the site area or at a ratio of 1:300 of GFA, whichever is larger. Greater than 80% of plants must be indigenous. Tree beer 250m2 site.	2	0	
	37	Nature connectivity	CA	Species Connectivity	Provide either landscaping or infrastructure to promote movement. Each conservation area must be at least 185m2. Connect to green/blue grid strategy where relevant for the site.	2	0	
	38	Nature Stewardship	CA	Nature Stewardship	Restore offsite biodiversity equivalent to the total GFA of the development, or site area, whichever is greater. Offsite area must be same ecological value.	2	0	
	39	Waterway protection	CA	Waterway protection	Demonstrate an annual average flow reduction (ML/yr) of 40% compared to pre-development levels and meets specified pollution targets.	2	0	
			EP	Waterway protection	Demonstrate an annual average flow reduction (ML/yr) of 80% compared to pre-development levels and meets specified pollution targets.	2	0	
					TOTAL for Nature	**14**	**0**	
LEADERSHIP	40	Market Transformation	CA	Market Transformation	The project demonstrates: How a building solution or process is considered leading in their targeted sector, nationally or globally; or That the technology or process is not commonly used within Australia's building industry; or globally, depending on the context of the innovation claimed.	5	0	
	41	Leadership Challenges	CA	Leadership Challenges	Meet the requirements of the following innovation challenges identified by the GBCA	5	0	
					TOTAL for Leadership	**10**	**0**	

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